COUGAR BATTERY POSITIVE CABLE INSPECTION

AFFECTED VEHICLES: **1999-2000 MODEL YEAR COUGAR VEHICLES EQUIPPED WITH 2.5L ENGINES**

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OVERVIEW

The affected vehicles in this program may have been built with misrouted battery positive cables or a loose connection. All cables that have been damaged due to these conditions must be replaced. This procedure will give the details for identifying misrouted cables, instruct you on how to properly route and retain the cable or replace the cable if damaged.

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INSPECTION

B+ TERMINAL AND CABLE

1. Install a memory saver, then disconnect the battery negative cable.
2. Raise the vehicle on a hoist.
3. Inspect the entire length of the battery positive cable and the connection at the B+ terminal as follows. See Figure 1.
   A. Check the cable insulation for chafing or gouging (possible contact with alternator or power steering line).
   B. Check the cable and rubber cover for signs of damage (burnt or melted).
   C. Lift up the rubber cover and check the B+ terminal for a loose connection or signs of arcing.
      - If any of the above damage is found, proceed to Battery Positive Cable Replacement.
      - If no damage is found, proceed to Battery Positive Cable Routing.

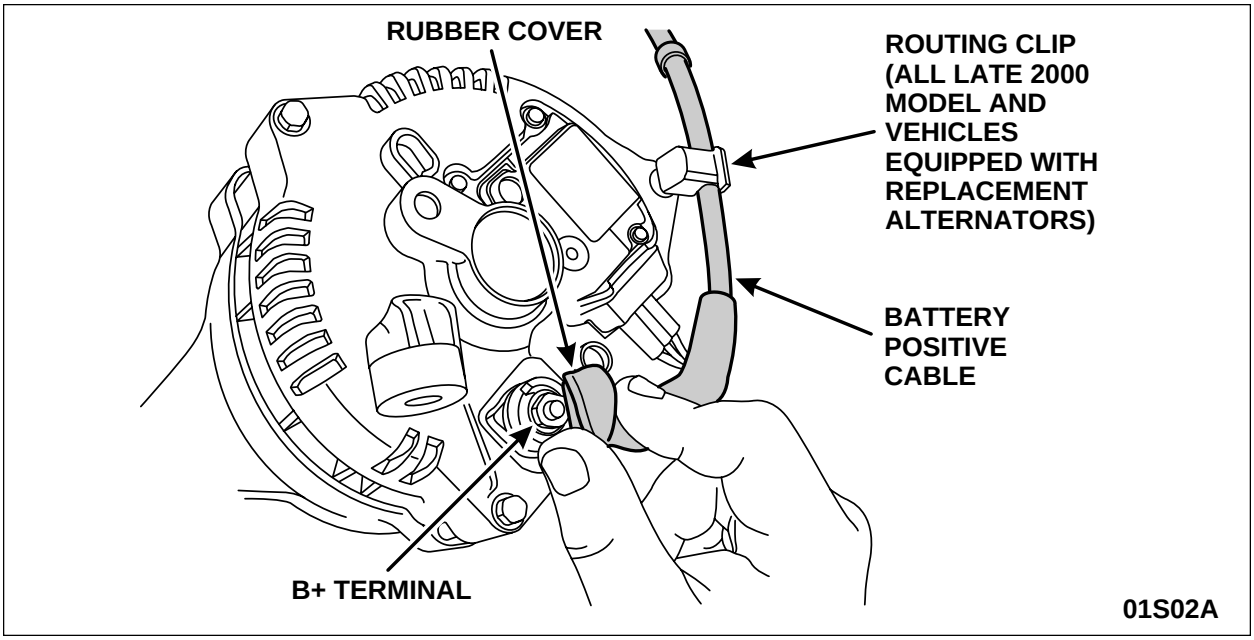


FIGURE 1

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BATTERY POSITIVE CABLE ROUTING

CAUTION: A routing clip mounting stud is located on the back of the alternator at the two (2) o'clock position on all late 2000 model vehicles. Some 1999 and early 2000 models may have this mounting stud if they are equipped with a replacement alternator. All vehicles with this mounting stud MUST have a routing clip installed on the stud.

1. NOTE: An improperly routed cable is depicted in Figure 2 for reference.

   Inspect the routing of the battery positive cable as follows. See Figures 3 and 4.

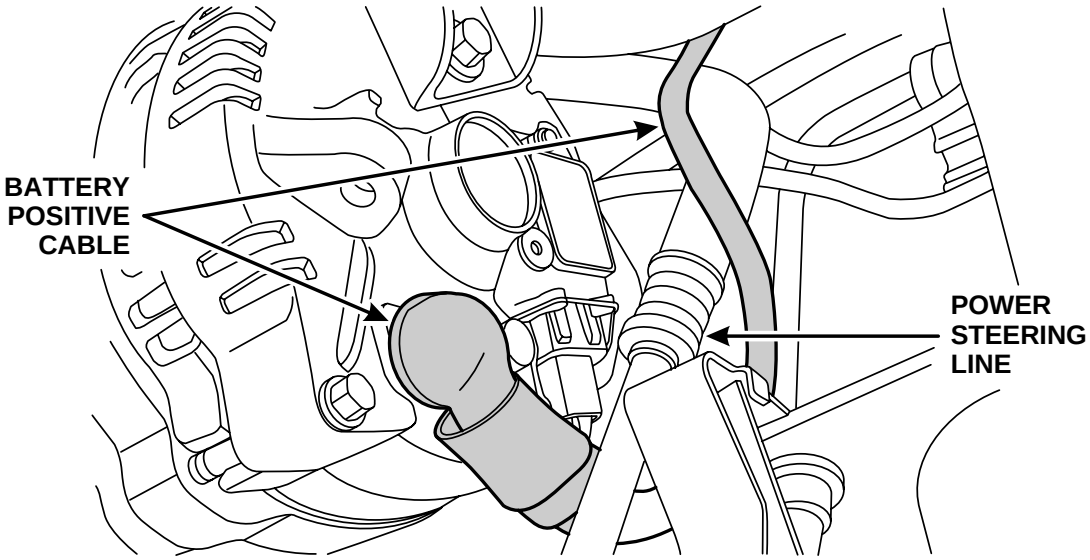
   A properly routed cable:
   - Extends up from the B+ terminal between the alternator and the power steering line, maintaining a minimum clearance of 19mm (3/4 inch) from the alternator and power steering line.
   - Passes through a routing clip attached to the back of the alternator at the 2 o'clock position (only if equipped with mounting stud).
   - Is secured to the alternator harness with a tie strap.

2. If the cable is properly routed, proceed as follows. See Figure 3 and 4.
   - Check for proper torque of the B+ terminal nut. The nut should be tightened to 8 Nm (71 lb-in).

   If the cable is properly routed except for a missing clip or tie strap, proceed as follows.
   - If missing, install a new clip (W702804-S300) to the back of the alternator (only if equipped with mounting stud) and attach the cable to the clip.
   - If missing, install a tie strap around the cable and alternator harness.
   - Check for proper torque of the B+ terminal nut. The nut should be tightened to 8 Nm (71 lb-in).

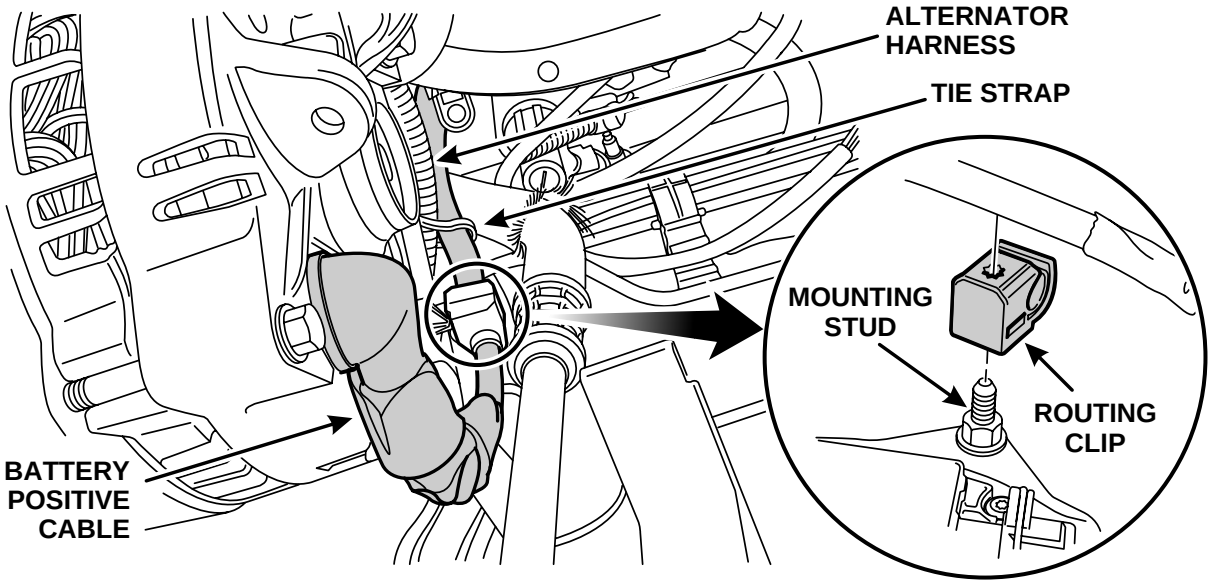
   If the cable is improperly routed, proceed as follows.
   - Disconnect the cable from the alternator and reroute the cable.
   - Connect the cable to the alternator and tighten the nut to 8 Nm (71 lb-in).
   - If missing, install a new clip (W702804-S300) to the back of the alternator (only if equipped with mounting stud) and attach the cable to the clip.
   - If missing, install a tie strap around the cable and alternator harness.

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IMPROPER ROUTING 01S02B

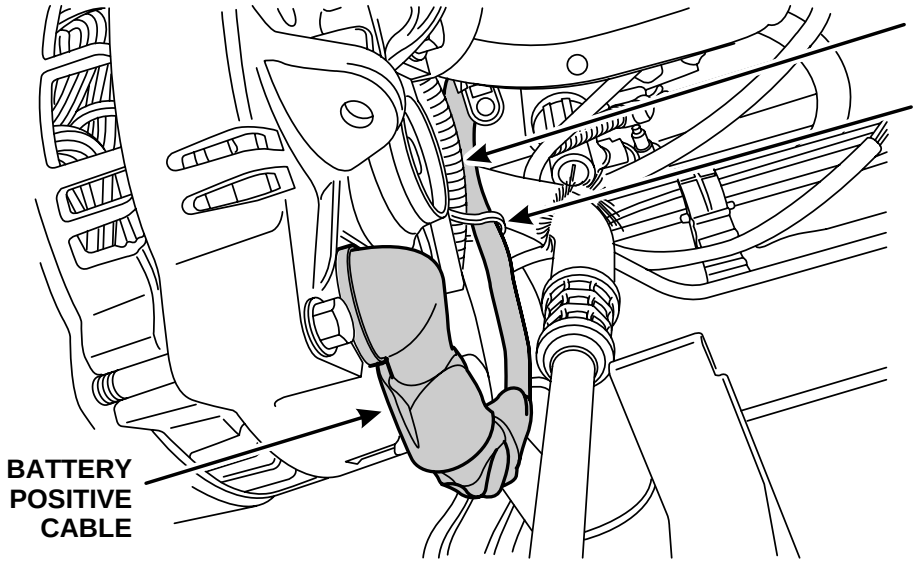
FIGURE 2



PROPER ROUTING – VEHICLES WITH ROUTING CLIP MOUNTING STUD 01S02C

FIGURE 3

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PROPER ROUTING – VEHICLES WITHOUT ROUTING CLIP MOUNTING STUD

01S02D

FIGURE 4

3. Lower the vehicle.
4. Connect the battery negative cable and remove the memory saver. No further inspection or repair is required.

BATTERY POSITIVE CABLE REPLACEMENT

REMOVAL

1. Lower the vehicle.
2. NOTE: The battery negative cable was disconnected in the inspection procedure above.

   Disconnect the power feed wire from the battery positive cable terminal.
3. Disconnect the battery positive cable and the starter solenoid inline connector.
4. Disconnect the intake air temperature (IAT) sensor and the mass air flow (MAF) sensor.
5. Disconnect the fresh air tubes from the air cleaner outlet tube.
6. Loosen the air cleaner outlet tube clamp at the throttle body, then remove the air cleaner assembly.
7. NOTE: A ground wire is secured by one of the ignition coil mounting bolts.

   Disconnect and remove the ignition coil.
8. Detach the battery positive cable from the routing clip on the battery tray.
9. Detach all of the pushpin tie straps securing the battery positive cable between the battery and the starter (if equipped).
10. Raise the vehicle.

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11. NOTE: The mega fuse holder release tab is most easily accessed from underneath the vehicle.

    Detach the two (2) tie straps and release the mega fuse holder from the engine control harness cover. See Figure 5.

12. Disconnect the battery positive cable from the B+ terminal, then release the cable from the routing clip (if equipped).

13. Detach the tie strap securing the battery positive cable to the alternator harness. See Figure 5.

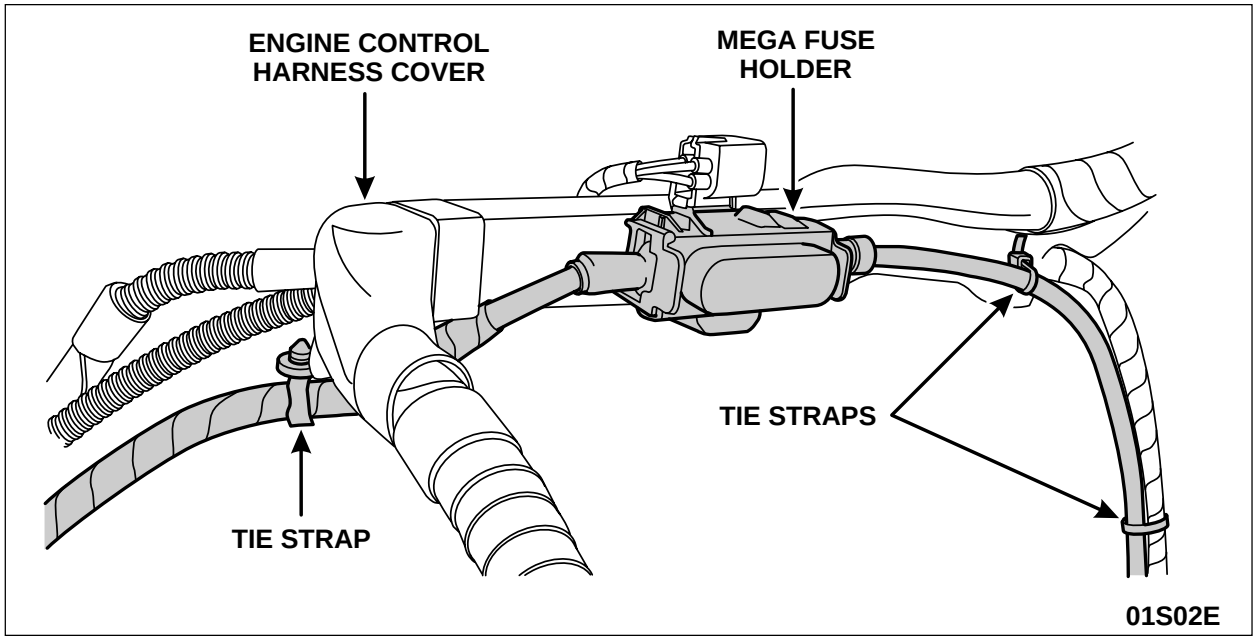


FIGURE 5

14. Lower the vehicle.

15. Disconnect the starter, then remove the battery positive cable from the vehicle.

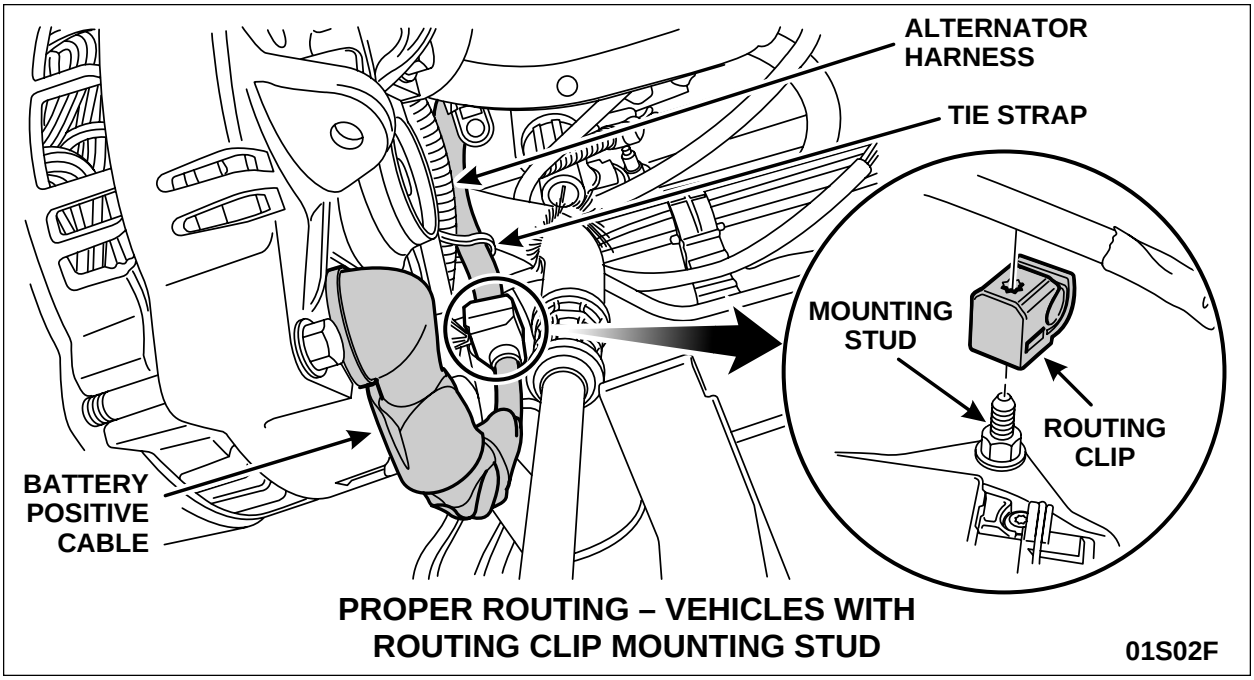
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INSTALLATION

1. Loosely route the new battery positive cable (XS8Z-14300-KB) on the engine.
2. Snap the mega fuse holder onto the engine control harness cover.
3. Raise the vehicle.
4. **CAUTION: A minimum of 19mm (3/4 inch) clearance must be maintained from the cable to the alternator and the power steering line.**

   Route the battery positive cable between the power steering line and the alternator, then connect the cable to the B+ terminal. Tighten the nut to 8 Nm (71 lb-in). See Figures 6 and 7.
5. **CAUTION: A routing clip mounting stud is located on the back of the alternator at the two (2) o'clock position on all late 2000 model vehicles. Some 1999 and early 2000 models may have this mounting stud if they are equipped with a replacement alternator. All vehicles with this mounting stud MUST have a routing clip installed on the stud.**

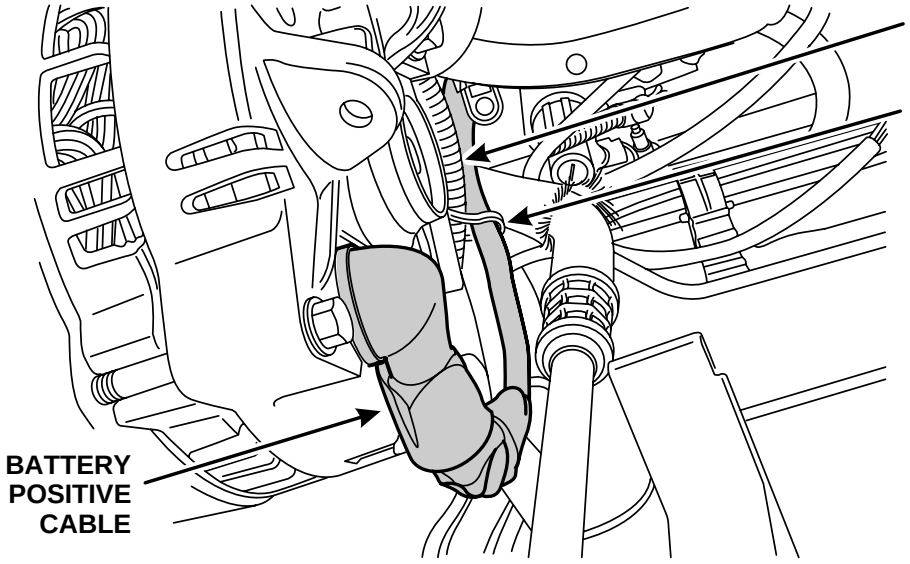
   The battery positive cable must pass through a routing clip mounted to the back of the alternator (only if equipped with mounting stud). If the routing clip is missing, a new one must be installed (W702804-S300). See Figure 6.
6. Secure the battery positive cable to the alternator harness with a tie strap. See Figures 6 and 7.



PROPER ROUTING – VEHICLES WITH ROUTING CLIP MOUNTING STUD **01S02F**

FIGURE 6

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PROPER ROUTING – VEHICLES WITHOUT ROUTING CLIP MOUNTING STUD

01S02G

FIGURE 7

7. Lower the vehicle.
8. Secure the battery positive cable to the engine control harness cover with two (2) tie straps.
9. NOTE: A ground wire is secured by one of the ignition coil mounting bolts.

   Install and connect the ignition coil. Tighten bolts to 6 Nm (53 lb-in).
10. Connect the starter. Tighten the large nut to 12 Nm (9 lb-ft) and the small nut to 5 Nm (44 lb-in).
11. Install all of the pushpin tie straps securing the battery positive cable between the battery and the starter (if equipped).
12. Attach the battery positive cable to the routing clip on the battery tray.
13. Install the air cleaner assembly, tighten the clamp on the air cleaner outlet tube securely.
14. Connect the MAF and the IAT.
15. Connect the battery positive cable and the starter solenoid inline connector.
16. Connect the power feed wire to the battery positive cable terminal. Tighten the nut to 8 Nm (71 lb-in).
17. Connect the battery negative cable, then remove the memory saver.

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